FunnyBirds: A Synthetic Vision Dataset for a Part-Based Analysis of Explainable AI Methods

Robin Hesse[1] • Simone Schaub-Meyer[1,2] • Stefan Roth[1,2]

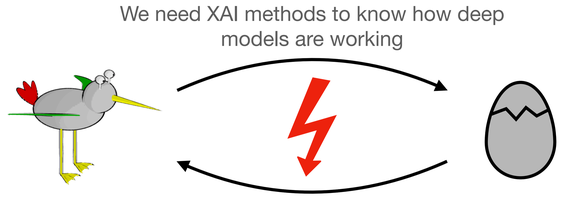
[1] TECHNISCHE UNIVERSITÄT DARMSTADT
[2] hessian.AI

ICCV23 PARIS

Paper & Code

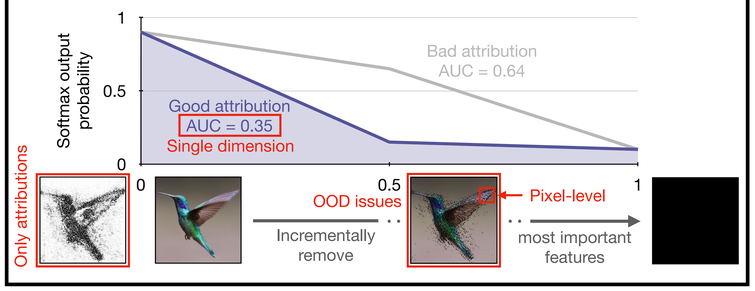
The ~~chicken~~ FunnyBird and egg problem

…of evaluating XAI methods



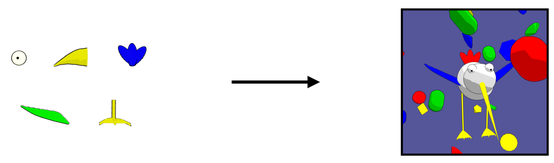
The feature deletion protocol e.g. Samek et al., 2017

…and its limitations



The FunnyBirds dataset

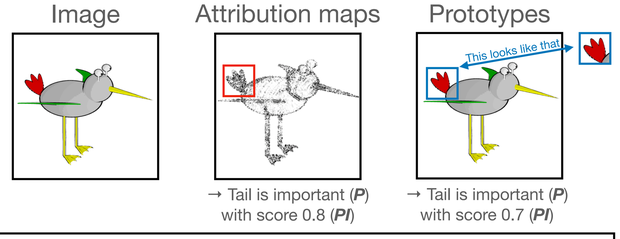
From concepts to FunnyBirds (50 classes)



→ Full control over / knowledge of the scene

Interface functions

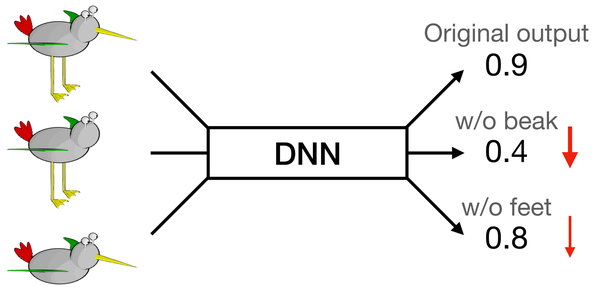
We evaluate different explanation types in a shared framework by implementing more general tasks.



→ Evaluation is build around part importances

The FunnyBirds framework

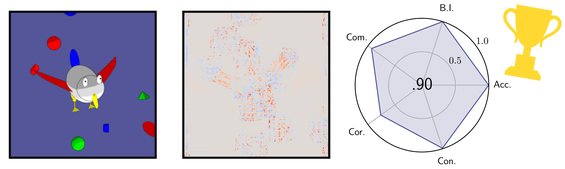
High-level idea: Perform image-space interventions to measure the output change and estimate "ground-truth" importances.



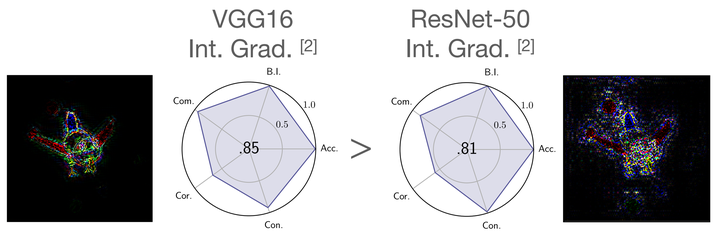
→ The beak is more important than the feet

Selected findings

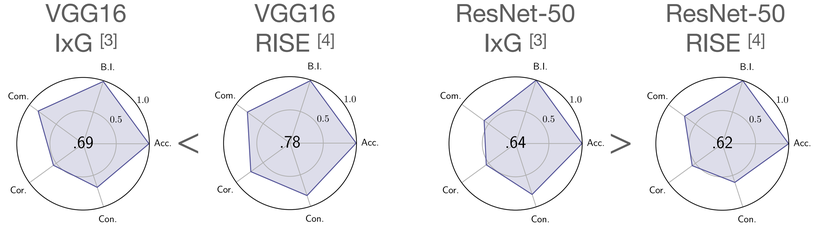
→ BagNet performs the best [1]



→ VGG16 is more explainable than ResNet-50



→ Rankings change across backbones



[1] Brendel & Bethge, 2019
[2] Sundararajan et al., 2017
[3] Shrikumar et al., 2017
[4] Petsiuk et al., 2018

This project has received funding from the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation programme (grant agreement No. 866008). The project has also been supported in part by the State of Hesse through the cluster projects "The Third Wave of Artificial Intelligence (3AI)" and "The Adaptive Mind (TAM)".

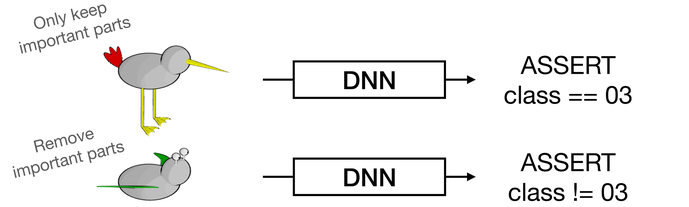
European Research Council
Established by the European Commission

Evaluation protocols

A multi-dimensional analysis framework for XAI methods.

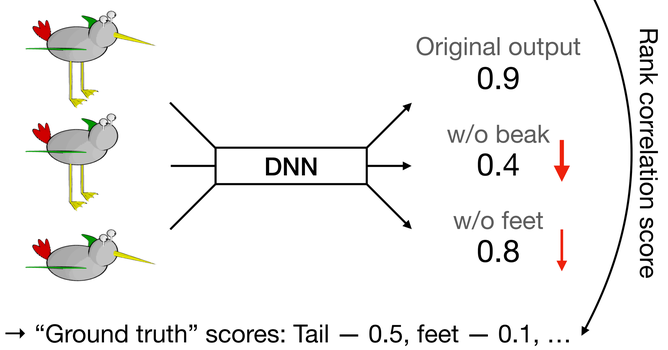
Completeness (Com.) measures the degree to which an explanation describes all aspects of the model's decision.

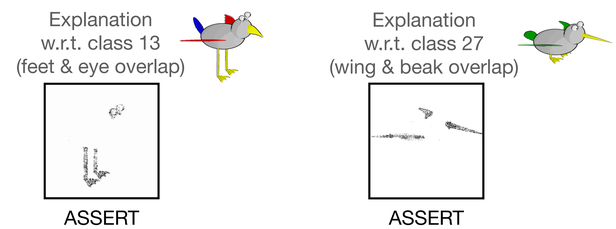
*Interface function (**P**): Tail, beak, and feet are important*



Correctness (Cor.) denotes the faithfulness of an explanation with respect to the model.

*Interface function (**PI**): Tail — 0.7, feet — 0.1, …*



→ "Ground truth" scores: Tail — 0.5, feet — 0.1, …

Contrastivity (Con.) measures the degree to which an explanation is discriminative w.r.t. different outputs.